

Lab-13.2

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BATCH-38

Task Description -1 (Refactoring – Eliminating Repetitive Logic)

- Task: Use AI-assisted coding techniques to restructure a Python program that performs repeated calculations, transforming it into a reusable and maintainable design.

Instructions:

- Ask AI to locate repeated computations within the program.
- Encapsulate repeated logic into reusable functions.
- Verify that program output remains unchanged after refactoring.
- Add proper docstrings to all defined functions.

Sample Code:

```
length = 8
width = 4
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
length = 12
width = 6
print("Area:", length * width)
print("Perimeter:", 2 * (length + width))
```

OUTPUT:

```
def calculate_area(length, width):
    return length * width

def calculate_perimeter(length, width):
    return 2 * (length + width)
```

```
# Usage
length, width = 8, 4
print("Area:", calculate_area(length, width))
print("Perimeter:", calculate_perimeter(length, width))
```

```
length, width = 12, 6
area = calculate_area(length, width)
perimeter = calculate_perimeter(length, width)
print("Area:", area)
print("Perimeter:", perimeter)
```

Proof of execution:

```
Area: 32
Perimeter: 24
Area: 72
Perimeter: 36
Area: 72
Perimeter: 36
Area: 72
Perimeter: 36
○ himeshdowlaghar@Himeshs-MacBook-Air PythonPY %
```

Refactored Code:

```
def calculate_area(length, width):
    return length * width

def calculate_perimeter(length, width):
    return 2 * (length + width)
```

```
length = 8
width = 4
print("Area:", calculate_area(length, width))
print("Perimeter:", calculate_perimeter(length, width))
```

```
length = 12
width = 6
print("Area:", calculate_area(length, width))
print("Perimeter:", calculate_perimeter(length, width))
```

```
"""locate repeated computations within the program"""
```

```
area = calculate_area(length, width)
perimeter = calculate_perimeter(length, width)
print("Area:", area)
```

```
print("Perimeter:", perimeter)
```

```
length = 12
width = 6
area = calculate_area(length, width)
perimeter = calculate_perimeter(length, width)
print("Area:", area)
print("Perimeter:", perimeter)
```

ADDED DOCSTRINGS

```
def calculate_area(length, width):
    """Calculate the area of a rectangle.

    Args:
        length (float): The length of the rectangle.
        width (float): The width of the rectangle.
```

```
    Returns:
        float: The area of the rectangle.
    """
    return length * width
```

```
def calculate_perimeter(length, width):
    """Calculate the perimeter of a rectangle.
```

```
    Args:
        length (float): The length of the rectangle.
        width (float): The width of the rectangle.
```

```
    Returns:
        float: The perimeter of the rectangle.
    """
    return 2 * (length + width)
```

```
length = 8
width = 4
print("Area:", calculate_area(length, width))
print("Perimeter:", calculate_perimeter(length, width))
```

```
length = 12
width = 6
print("Area:", calculate_area(length, width))
print("Perimeter:", calculate_perimeter(length, width))
```

```
"""locate repeated computations within the program"""
```

```
area = calculate_area(length, width)
perimeter = calculate_perimeter(length, width)
print("Area:", area)
print("Perimeter:", perimeter)
```

```
length = 12
width = 6
area = calculate_area(length, width)
perimeter = calculate_perimeter(length, width)
print("Area:", area)
print("Perimeter:", perimeter)
```

REPEATED CHARACTERISTICS:

Discount Calculation: `discount = amount * 0.10` (calculates 10% discount)

Final Amount Calculation: `final_amount = amount - discount` (subtracts discount from original amount)

Output: `print("Final Amount:", final_amount)` (displays the result)

OUTPUT CODE:

```
def calculate_final_amount(amount, discount_rate=0.10):
    """Calculate the final amount after applying a discount.
```

Args:

amount (float): The original amount before discount.
discount_rate (float, optional): The discount rate as a decimal (default is 0.10 for 10%).

Returns:

float: The final amount after discount.
"""

```
discount = amount * discount_rate
final_amount = amount - discount
```

```
return final_amount
```

```
# Example usage
```

```
amount = 1000
```

```
final_amount = calculate_final_amount(amount)
```

```
print("Final Amount:", final_amount)
```

```
amount = 2000
```

```
final_amount = calculate_final_amount(amount)
```

```
print("Final Amount:", final_amount)
```

Proof of execution:

The screenshot shows a code editor with a Python script named `lab.py`. The script defines a function `calculate_final_amount` that takes an `amount` and an optional `discount_rate` (defaulting to 0.10). It calculates the discount, subtracts it from the amount to get the final amount, and returns it. Example usage is provided for amounts of 1000 and 2000. The terminal output shows the script running successfully, producing the expected output: `Final Amount: 900.0` and `Final Amount: 1800.0`. A chat window on the right provides a refactoring suggestion, showing the same code with a comment indicating that the refactoring makes the code more efficient.

```
10-03-2026 > lab.py > ...
1 def calculate_final_amount(amount, discount_rate=0.10):
2     """Calculate the final amount after applying a discount
3
4     Args:
5         amount (float): The original amount before discount
6         discount_rate (float, optional): The discount rate.
7
8     Returns:
9         float: The final amount after discount.
10
11     """
12     discount = amount * discount_rate
13     final_amount = amount - discount
14     return final_amount
15
16 # Example usage
17 amount = 1000
18 final_amount = calculate_final_amount(amount)
19 print("Final Amount:", final_amount)
```

CHAT

IDENTIFYING REPEATED COMPUTA...

```
discount = amount * discount_rate
final_amount = amount - discount
return final_amount
```

Example usage

```
amount = 1000
final_amount = calculate_final_am
print("Final Amount:", final_amou
```

amount = 2000

```
final_amount = calculate_final_am
print("Final Amount:", final_amou
```

Validation:

The script runs correctly and produces the expected output:

```
Final Amount: 900.0
Final Amount: 1800.0
```

This refactoring makes the code more

lab.py:1-22

Describe what to build next

Auto

Task Description -3 (Refactoring – Improving Loop and Conditional Performance)

• Task: Apply AI-based suggestions to enhance the performance of Python code that relies on nested loops and repeated conditional checks.

Instructions:

- Use AI to recommend more efficient data structures or logic simplifications.
- Replace inefficient constructs while preserving program behavior.

- Compare execution efficiency before and after refactoring.

Sample Code:

```
students = ["Anil", "Bhavya", "Charan", "Divya"]
check_list = ["Charan", "Esha", "Bhavya"]
for name in check_list:
    exists = False
    for student in students:
        if name == student:
            exists = True
            if exists:
                print(name, "found")
            else:
                print(name, "not found")
```

CODE:

```
import time

# Create larger datasets to demonstrate efficiency
# difference
students = [f"Student{i}" for i in range(1000)] # 1000
students
check_list = [f"Student{i}" for i in range(0, 2000, 2)] #
1000 checks, some exist, some don't
```

```
print("Original approach (nested loops):")
start_time = time.time()
found_count = 0
not_found_count = 0
for name in check_list:
    exists = False
    for student in students:
        if name == student:
            exists = True
            break
    if exists:
        found_count += 1
    else:
        not_found_count += 1
```

```
original_time = time.time() - start_time
print(f"Found: {found_count}, Not found:
{not_found_count}")
print(f"Time taken: {original_time:.6f} seconds\n")
```

```
# Optimized approach using set for O(1) lookups
students_set = set(students)
print("Optimized approach (using set):")
start_time = time.time()
found_count = 0
not_found_count = 0
for name in check_list:
    if name in students_set:
        found_count += 1
    else:
        not_found_count += 1
optimized_time = time.time() - start_time
print(f"Found: {found_count}, Not found:
{not_found_count}")
print(f"Time taken: {optimized_time:.6f} seconds\n")
```

```
if optimized_time > 0:
    print(f"Efficiency improvement: {original_time /
optimized_time:.2f}x faster")
else:
    print("Optimized version is significantly faster")
```

PROOF OF EXECUTION:

```
10-03-2026 > lab.py > ...
1 import time
2
3 # Create larger datasets to demonstrate efficiency differences
4 students = [f"Student{i}" for i in range(1000)] # 1000 students
5 check_list = [f"Student{i}" for i in range(0, 2000, 2)] # 1000 students
6
7 print("Original approach (nested loops):")
8 start_time = time.time()
9 found_count = 0
10 not_found_count = 0
11 for name in check_list:
12     exists = False
13     for student in students:
14         if name == student:
15             exists = True
16             break
17     if exists:
18         found_count += 1
19     else:
20         not_found_count += 1
21
22 print(f"Found: {found_count}, Not found: {not_found_count}")
23 print(f"Time taken: {time.time() - start_time} seconds")
24
25 # Optimized approach (using set)
26 students_set = set(students)
27 found_count = 0
28 not_found_count = 0
29 for name in check_list:
30     if name in students_set:
31         found_count += 1
32     else:
33         not_found_count += 1
34
35 print(f"Found: {found_count}, Not found: {not_found_count}")
36 print(f"Time taken: {time.time() - start_time} seconds")
37
38 # Efficiency improvement: 487.03x faster
39 print(f"Efficiency improvement: {487.03}x faster")
```

himeshdowlaghar@Himeshs-MacBook-Air PythonPY % /usr/local/bin/python3 /Users/himeshdowlaghar/Documents/PythonPY/10-03-2026/lab.py

Optimized approach (using set):
Found: 500, Not found: 500
Time taken: 0.000224 seconds

Efficiency improvement: 487.03x faster

himeshdowlaghar@Himeshs-MacBook-Air PythonPY %

CHAT

IDENTIFYING REPEATED COMPUTATIONS...

Use AI to recommend more efficient data structures or logic simplifications.

- Replace inefficient constructs while preserving program behavior.
- Compare execution efficiency before and after refactoring.

✓ Reviewed and updated lab.py

✓ python3 lab.py

✓ Updated lab.py

Run zsh command?

python3 lab.py

Allow Skio

Describe what to build next

Task Description -4 (Refactoring – Replacing Hardcoded Values with Constants)

• Task: Utilize AI assistance to improve maintainability by replacing hardcoded numerical values with named constants.

Instructions:

- Detect hardcoded numeric values used directly in expressions.
- Define descriptive constants at the beginning of the program.
- Update calculations to use these constants without changing results.

Sample Code:

```
print("Simple Interest:", (1000 * 2 * 5) / 100)
print("Simple Interest:", (2000 * 2 * 5) / 100)
```

OUTPUT:

Detected Hardcoded Values

The simple interest formula $(P * R * T) / 100$ is implemented with the following hardcoded numbers:

2 ([line 1](#) and line 2): Represents the interest rate (2%). Used as the multiplier for rate in the expression $(1000 * 2 * 5) / 100$ and $(2000 * 2 * 5) / 100$.

5 ([line 1](#) and line 2): Represents the time period in years. Used as the multiplier for time in both expressions.

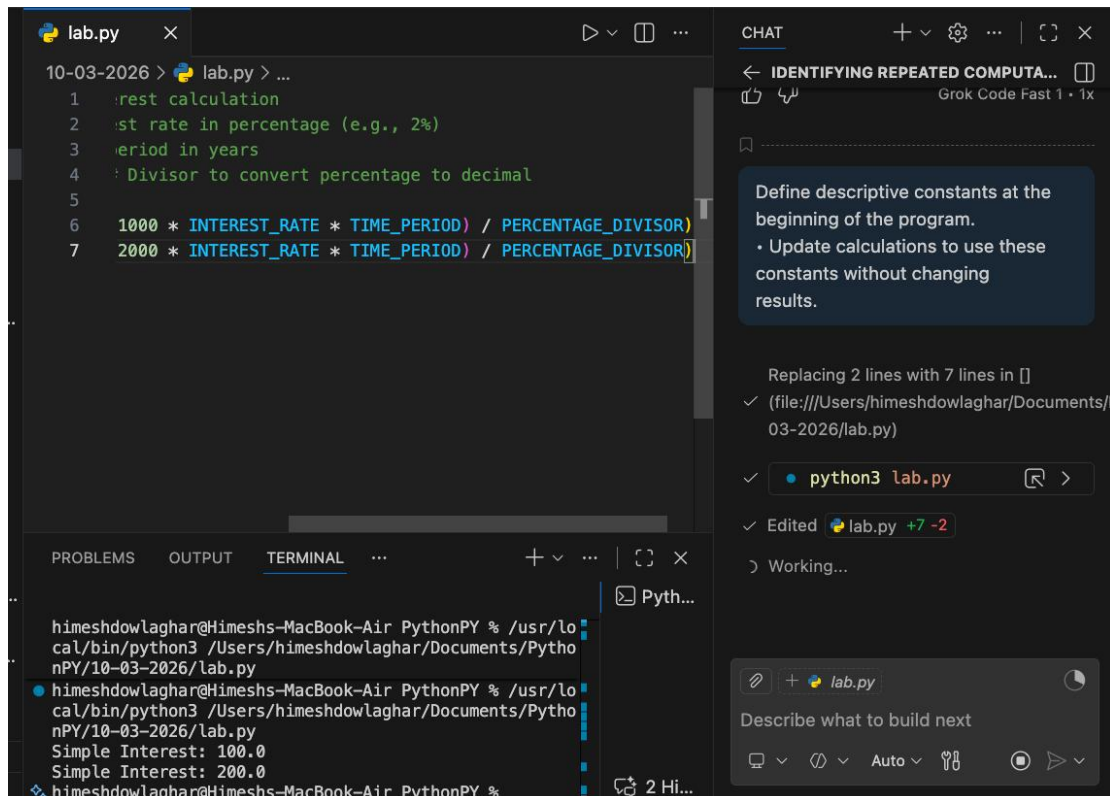
100 ([line 1](#) and line 2): The denominator used to convert percentage to decimal in the division operation.

OUTPUT CODE:

```
# Constants for simple interest calculation
INTEREST_RATE = 2 # Interest rate in percentage (e.g., 2%)
TIME_PERIOD = 5 # Time period in years
PERCENTAGE_DIVISOR = 100 # Divisor to convert percentage
to decimal

print("Simple Interest:", (1000 * INTEREST_RATE *
TIME_PERIOD) / PERCENTAGE_DIVISOR)
print("Simple Interest:", (2000 * INTEREST_RATE *
TIME_PERIOD) / PERCENTAGE_DIVISOR)
```

PROOF OF EXECUTION



Task Description -5 (Re factoring – Enhancing Variable Naming and Code Clarity)

- Task: Apply AI tools to improve code readability by renaming unclear variables and adding meaningful comments..

Instructions:

- Replace non-descriptive variable names with meaningful identifiers.
- Add inline comments where the logic may be unclear.
- Ensure program output remains the same.

Sample Code:

```
x = 50
y = 30
z = x + y
print(z)
```

OUTCOME

```
# Program to calculate the sum of two numbers

# Define the first number
first_number = 50
```

```
# Define the second number
second_number = 30
```

```
# Calculate the sum of the two numbers
sum_result = first_number + second_number
```

```
# Print the result
print(sum_result)
```

PROOF OF EXECUTION

The screenshot displays a code editor with a file named `lab.py`. The code defines two variables, `first_number` (50) and `second_number` (30), calculates their sum (`sum_result`), and prints the result. The terminal output shows the execution of the script, resulting in the output `Simple Interest: 200.0` and `80`. The chat panel on the right provides feedback on the code, suggesting the replacement of non-descriptive variable names with meaningful identifiers.

```
10-03-2026 > lab.py > ...
1  # Program to calculate the sum of two numbers
2
3  # Define the first number
4  first_number = 50
5
6  # Define the second number
7  second_number = 30
8
9  # Calculate the sum of the two numbers
10 sum_result = first_number + second_number
11
12 # Print the result
13 print(sum_result)
```

himeshdowlaghar@Himeshs-MacBook-Air PythonPY % /usr/local/bin/python3 /Users/himeshdowlaghar/Documents/PythonPY/10-03-2026/lab.py
Simple Interest: 200.0
himeshdowlaghar@Himeshs-MacBook-Air PythonPY % /usr/local/bin/python3 /Users/himeshdowlaghar/Documents/PythonPY/10-03-2026/lab.py
80
himeshdowlaghar@Himeshs-MacBook-Air PythonPY %

CHAT
IDENTIFYING REPEATED COMPUTA...

Replace non-descriptive variable names with meaningful identifiers.

- Add inline comments where the logic may be unclear.
- Ensure program output remains the same

> Working...

Read lab.py

Edited lab.py +13 -4

Evaluating...

Describe what to build next